# 2.1 Measure Preliminaries-1

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* Let $X$ be a non empty set. An algebra of sets of $X$ is a non-empty collection $\mathcal{A}$ of subsets of $X$ closed under finite union and complement.

A $\sigma$-algebra on $X$ is a collection $\mathcal{M}$ of subsets of $X$ satisfying

(ci) $\emptyset \in \mathcal{M}$

(ii) If $E \in \mathcal{M}$ then $E^c = X \setminus E \in \mathcal{M}$.

(ciii) If $\{E_n\}_{n\in\mathbb{N}}$ is a seq$^n$ of sets in $\mathcal{M}$, then $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$.

Now $(X, \mathcal{M})$ is a measurable space and elements of $\mathcal{M}$ are called measurable set.

Properties of $\sigma$-algebra.

1) $X \in \mathcal{M}$

2) If $E_1, \ldots, E_N \in \mathcal{M}$ then $\bigcup_{i=1}^{N} E_i \in \mathcal{M}$

3) If $E_1, \ldots, E_N \in \mathcal{M}$ then $\bigcap_{i=1}^{N} E_i \in \mathcal{M}$

4) If $\{E_i\}_{i\in\mathbb{N}}$ is a seq$^n$ of sets in $\mathcal{M}$, then $\bigcap_{i=1}^{\infty} E_i \in \mathcal{M}$

5) If $E, F \in \mathcal{M}$ then $E \setminus F \in \mathcal{M}$.

Proof:

1) $\emptyset \in \mathcal{M}$ and $X = (\emptyset)^c \in \mathcal{M}$.

2) Let $\{E_i\}_{i\in\mathbb{N}}$ be a countable collection of measurable sets where $E_i = \emptyset$ for $i > N$. Hence, $\bigcup_{i=1}^{\infty} E_i = E_1 \cup \cdots \cup E_N \in \mathcal{M}$.

4) $(\bigcup_{i=1}^{\infty} E_i)^c = \bigcap_{i=1}^{\infty} E_i^c \in \mathcal{M}$

3) Apply 4) to collection of sets in 2)

5) $E \setminus F = E \cap F^c \in \mathcal{M}$ as finite intersection

is closed under $\sigma$-algebra.

**Proposition**

If $\{M_\alpha \mid \alpha \in J\}$ is collection of $\sigma$-algebra on $X$. Then $\bigcap_{\alpha\in J} M_\alpha$ is a $\sigma$-algebra.

**Proof:**

Observe $\emptyset \in M_\alpha$ for all $\alpha \in J$ this implies $\emptyset \in \bigcap_{\alpha\in J} M_\alpha$. If $E \in \bigcap_{\alpha\in J} M_\alpha$ then $E \in M_\alpha$ for all $\alpha \in J$ hence $E^c \in M_\alpha$ for all $\alpha \in J$ this implies $E^c \in \bigcap_{\alpha\in J} M_\alpha$. If $\{E_n\}_{n\in\mathbb{N}}$ is a seq$^n$ of sets in $\bigcap_{\alpha\in J} M_\alpha$ hence $\bigcup_{n\in\mathbb{N}} E_n \in M_\alpha$ for each $\alpha \in J$. Hence $\bigcup_{n\in\mathbb{N}} E_n \in \bigcap_{\alpha\in J} M_\alpha$.

$\square$

* In particular let $S \subseteq \mathcal{P}(X)$, there exists a unique smallest $\sigma$-algebra on $X$ that contains $S$. This is called $\sigma$-algebra generated by $S$.

*Remark* 0.1. We show uniqueness by constructing a collection of $\sigma$-algebra containing $S$ and then taking an intersection over the family of $\sigma$-algebras. By the above proposition, the resulting collection is the smallest $\sigma$-algebra containing $S$.

* Let $X$ be a topological space. The $\sigma$-algebra generated by topology on $X$ is called Borel $\sigma$-algebra on $X$ and denoted by $B_X$. The members of the $\sigma$-algebra are called Borel sets.

Note: $F_\sigma$ is countable union of closed sets and $G_\delta$ is countable intersection of open sets. These are examples of Borel sets.

Proposition

$\mathcal{B}_\mathbb{R}$ is generated by each where $\mathcal{B}_{1\mathbb{R}}$ is borel $\sigma$-algebra.

a. open intervals $\mathcal{E}_1$

b. closed intervals $\mathcal{E}_2$

c. half-open intervals $\mathcal{E}_3$

d. open rays $\mathcal{E}_4$

e. closed rays $\mathcal{E}_5$

Proof: Observe $\mathcal{E}_1$, $\mathcal{E}_2$, $\mathcal{E}_4$ and $\mathcal{E}_5$ are open or closed sets. For elements $\mathcal{E}_3$, these are $G_\delta$ sets i.e. $(a,b] = \bigcap_{n=1}^\infty (a, b+\frac{1}{n})$. Hence all are borel sets. If $\mathcal{M}(\mathcal{E}_j)$ is $\sigma$-algebra generated by $\mathcal{E}_j$. Then $\mathcal{M}(\mathcal{E}_j) \subseteq \mathcal{B}_{1\mathbb{R}}$ for $j = 1,2,3,4,5$.

As any open set in $\mathbb{R}$ can be expressed as union of countable open intervals. Hence $B_\mathbb{R} \subseteq M(\mathcal{E})$.

Now observe for any open interval $(a,b)$

$$(a,b) = \bigcap_{n=1}^{\infty}\left[a-\frac{1}{n}, b+\frac{1}{n}\right] = \left(a+\frac{1}{n}\right) \cap \left[a-\frac{1}{n}, b\right)$$

$$= (a,\infty) \cap \bigcap_{n=1}^{\infty}\left(-\infty, b+\frac{1}{n}\right)$$

$$= \bigcap_{n=1}^{\infty}\left[a-\frac{1}{n}, \infty\right) \cap (-\infty, b)$$

Hence $M(\mathcal{E}_j) = B_\mathbb{R}$ for $j = 1,2,3,4,5$.

**Bonus!** Every open set in $\mathbb{R}$ is a countable union of open intervals.

Let $U$ be open set in $\mathbb{R}$. If $x \in U$ then $x$ is rational or irrational. Let $x$ be rational then construct $I_x = \bigcup_{x\in I} I$ where $I$ are non disjoint open intervals containing $x$ and contained in $U$. If $x$ is not a rational for some $\epsilon > 0$, $x \in (a-\epsilon, a+\epsilon) \subseteq U$ choose a rational $q \in (a-\epsilon, a+\epsilon)$ and observe

$x \in I_q$. Hence for $x \in U$ there exists $q \in \mathbb{Q}$ st. $x \in I_q$. This implies $U \subseteq \bigcup_{q\in\mathbb{Q}} I_q$. Observe $I_q \subseteq U$ for all $q \in \mathbb{Q}$ hence

$$U = \bigcup_{q\in\mathbb{Q}} I_q \text{ where } I_q \text{ are open intervals.}$$

* A family of set $\mathcal{R} \subseteq \mathcal{P}(X)$ is called a ring if it closed under finite union and difference (i.e. if $E_1, \ldots, E_n \in \mathcal{R}$ then $\bigcup_{i=1}^{n} E_i \in \mathcal{R}$ and if $E, F \in \mathcal{R}$ then $E \setminus F \in \mathcal{R}$). A ring that is closed under countable unions is called a $\sigma$-ring.

**Proposition**

If $\mathcal{R}$ is a $\sigma$-ring are closed under countable intersection.

**Proof:** For $F \in \mathcal{R}$ observe $\emptyset = F \setminus F \in \mathcal{R}$. Let $\{E_i\}_{i=1}^{\infty}$ be a collection of sets belonging

to $R$. If any of them are pairwise disjoint $\bigwedge_{i=1}^{n} E_i = \emptyset \in R$. Assume $\{E_i\}_{i=1}^{n}$ are not pairwise disjoint. As $R$ is a $\sigma$-ring, let $\bigcup_{i=1}^{n} E_i = E \in R$. Let $E_i' = E \setminus E_i \in R$ for $i \in \mathbb{N}$. Now $E' = \bigcup_{i=1}^{n} E_i' \in R$. Observe $\bigwedge_{i=1}^{n} E_i = E \setminus E' \in R$.

* Let $X$ be a set equipped with $\sigma$-algebra $\mathfrak{M}$

We define countably additive measure $m$ such that

(i) $m : \mathfrak{M} \to [0, \infty]$ is well defined function

(ii) $m(\emptyset) = 0$.

(iii) If $\{E_j\}_{j=1}^{\infty}$ is a seq$^n$ of disjoint sets then

$$m\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} m(E_j)$$

Observe (iii) implies finite additivity by taking $E_j = \emptyset$ for $j > n$.

* We define $(X, \mathfrak{M}, m)$ as a measure space.

If $m(X) < +\infty$ we call $m$ as finite measure

If $X = \bigcup_{j=1}^{\infty} E_j$ where $E_j \in \mathfrak{M}$ and $m(E_j) < +\infty$ for all $j$ then $m$ is $\sigma$-finite measure.

More generally $E$ is $\sigma$-finite set if $E = \bigcup_{j=1}^{\infty} E_j$

$E_j \in \mathfrak{M}$ and $m(E_j) < +\infty$ for all $j$.

* If for each $E \in \mathcal{M}$ with $m(E) = \infty$ and there exists $F \subseteq E$ st. $0 < m(F) < \infty$ then $m$ is semifinite.

**Exercise**

Every $\sigma$-finite measure is semifinite.

**Proof:**

If $m(E) = \infty$ then observe $E = E \cap X = \bigcup_{j=1}^{\infty} (E \cap E_j)$ this implies $E \cap E_j \subseteq E_j \subseteq X$ for some $j$. We know $m(E_j) < +\infty$ hence $m(E \cap E_j) < m(E_j) < +\infty$. Hence we have found a subset of $E$ specifically $E \cap E_j$ which has measure of $E_j \cap E$ to be finite.

*Example* 0.1. (i) Let $X$ be any set and $x_0 \in X$ be a fixed point and $M = \mathcal{P}(X)$. Let $m : M \to \mathbb{R}$

$$m(E) = \begin{cases} 1 & \text{if } x_0 \in E \\ 0 & \text{if } x_0 \notin E \end{cases}$$

This is called the Dirac measure.

(ii) Let $X$ be any set and $M = \mathcal{P}(X)$. Let $m : M \to [0, \infty]$

$$m(E) = \begin{cases} |E| & \text{if } |E| < +\infty \\ \infty & \text{otherwise} \end{cases}$$

where $|\cdot|$ is cardinality function and $m$ is called the counting measure.

(iii) Let $X$ be any set and $M = \mathcal{P}(X)$. Let $m : M \to [0, \infty]$

$$m(E) = +\infty \text{ for all } E \subset \mathbb{R}$$

**Lemma 0.1.** Suppose $X$ be a set and $\mathfrak{M}$ is a $\sigma$-algebra equipped on $X$. Let $m : \mathfrak{M} \to [0, \infty]$ be a measure, then

(i) [Monotonicity] : If $A, B \in \mathfrak{M}$ s.t. $A \subseteq B$ then

$$m(CA) \leq m(B).$$

(ii) If $\{A_n\} \subseteq \mathfrak{M}$ then

$$m\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leq \sum_{n\in\mathbb{N}} m(A_n).$$

*Proof.* (i) If $B \supseteq A$ then $m(B) = m(A \cup B \setminus A)$

$$= m(A) + m(B \setminus A) \geq m(A).$$

(ii) Let $E_1 = A_1$, and $E_k = A_k \setminus \bigcup_{i=1}^{k-1} A_i$ for $k \geq 1$. Then observe all $E_k$ are disjoint to each other hence

$$m\left(\bigcup_{k\in\mathbb{N}} E_k\right) = \sum_{k\in\mathbb{N}} m(E_k)$$

then note

$$\bigcup_{n\in\mathbb{N}} A_n = \bigcup_{k\in\mathbb{N}} E_k$$

and $m(E_k) \leq m(A_k)$ for all $k \in \mathbb{N}$. Hence we get

$$m\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leq \sum_{k\in\mathbb{N}} m(A_k).$$

$\square$

**Lemma 0.2.** Let $(X, \mathcal{M}, \mu)$ be a measure space.

(a) Continuity from below: If $\exists E_j \uparrow_j^\infty = 1 \subset \mathcal{M}$ and $E_1 \subset E_2 \subset \cdots$, then $\mu(\bigcup_j E_j) = \lim_{j\to\infty} \mu(E_j)$.

(b) Continuity from above: If $\exists E_j \downarrow_j^\infty = 1 \subset \mathcal{M}$ and $E_1 \supset E_2 \supset E_3 \supset \cdots$, then $\mu(\bigcap_j E_j) = \lim_{j\to\infty} \mu(E_j)$. and $\mu(E_1) < +\infty$.

*Proof.* (a) Set $E_0 = \emptyset$. Observe $\bigcup_j E_j = \bigcup_{j=1}^{\infty} E_j \setminus E_{j-1}$. Hence, $\mu(\bigcup_j E_j) = \mu(\bigcup_{j=1}^{\infty} E_j \setminus E_{j-1})$
$= \sum_{j=1}^{\infty} \mu(E_j \setminus E_{j-1}) = \lim_{n\to\infty} \sum_{j=1}^{n} \mu(E_j \setminus E_{j-1})$
$= \lim_{n\to\infty} \mu(E_n)$.
(b) Let $F_j = E_1 \setminus E_j$. Observe $F_1 \subset F_2 \subset \cdots$ and $\bigcup_{j=1}^{\infty} F_j = E_1 \setminus \bigcap_j E_j$.
$\mu(E_1) = \mu(\bigcap_j E_j) + \lim_{n\to\infty} \mu(F_n)$ □

$$\mu(E_i) = \mu\left(\bigcap_{i=1}^{\infty} E_j\right) + \lim_{n\to\infty} \mu(E_i \setminus E_j)$$

$$= \mu\left(\bigcap_{j=1}^{\infty} E_j\right) + \mu(E_i) - \lim_{n\to\infty} \mu(E_j).$$

This implies $\mu\left(\bigcap_{i=1}^{\infty} E_j\right) = \lim_{n\to\infty} \mu(E_j)$, as $\mu(E_i) < +\infty$, we can subtract it from both sides.

□

* If $(X, \mathcal{M}, \mu)$ is a measure space, a set $E \in \mathcal{M}$ such that $\mu(CE) = 0$ is called a null set.

If a statement is true about point $x \in X \setminus E$ for some null set $E$ then we say the statement holds "almost everywhere" (a.e) (More precisely, we have $\mu$-Null set and $\mu$-a.e).

**Corollary**

Any countable union of null sets is a null set, and subsets of null sets belonging in $\mathcal{M}$ are null sets.

* If a measure whose domain include all subsets of a null set then the domain is called complete.
**Theorem 0.1.** Suppose that $(X, \mathcal{M}, \mu)$ is a measure space. Let $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$ and $\overline{\mathcal{M}} = \{E \cup F \mid E \in \mathcal{M} \text{ and } F \subset N \text{ for some } N \in \mathcal{N}\}$. Then $\overline{\mathcal{M}}$ is a $\sigma$-algebra and there is a unique extension $\overline{\mu}$ of $\mu$ to complete measure $\overline{\mathcal{M}}$.

*Proof.* (a) $\overline{\mathcal{M}}$ is a $\sigma$-algebra

Since $\mathcal{M}$ and $\mathcal{N}$ are closed under countable unions, so is $\overline{\mathcal{M}}$. If $E \in \mathcal{M}$ and $F \subset N \in \mathcal{N}$ then $E \cup F \in \overline{\mathcal{M}}$. Now assume $E \cap N = \emptyset$ otherwise replace $F$ by $F \setminus E$ and $N$ by $N \setminus E$. Then $E \cup F = (E \cup N) \cap (N^c \cup F)$, so $(E \cup F)^c = (E \cup N)^c \cup (N^c \cup F)^c$ and observe

$(EUN)^c \in \mathcal{M}$ and $N \setminus F^c = N \setminus F \subset N$ and $N \setminus F \in \mathcal{N}$. Hence $(EUF)^c \in \overline{\mathcal{M}}$.

(cb) Define $\overline{\mu} : \overline{\mathcal{M}} \to [0, \infty]$, such that $\overline{\mu}(EUF) = \mu(CE)$. We want to show well definedness. Let $E_1UF_1 = E_2UF_2$ where $E_1, E_2 \in \mathcal{M}$ and $F_1, F_2 \in \mathcal{N}$. Now, $E_1 \subset E_1UF_1 = E_2UF_2 \subset E_2UN_2$. This implies $\mu(CE_1) \leq \mu(E_2) + \mu(N_2) = \mu(E_2)$. If $E_2 \subset E_1UN$, and this implies $\mu(E_2) \leq \mu(E_1)$. Hence $\mu(CE_1) = \mu(CE_2)$.

(c) Uniqueness. Let $m_1, m_2$ be extension of $\mu$ as per the above extension. Now, $EUF \in \overline{\mathcal{M}}$. Observe $m_1(EUF) = \mu(CE) = m_2(EUF')$ where $F'$ is any subset of a null set.

* Outer measure

* An outer measure on a non-empty set $X$ is a function $\mu^* : \mathcal{P}(X) \to [0, \infty]$ that satisfy,

* $\mu^*(\emptyset) = 0$.

* $\mu^*(A) \leq \mu^*(B)$ if $A \subseteq B$.

* $\mu^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$

**Proposition**

Let $E \subseteq \mathcal{P}(X)$ and $p : E \to \mathcal{P}(X)$ and $p : E \to [0, \infty]$ be such that $\emptyset \in E$, $X \in E$ and $p(\emptyset) = 0$. For any $A \subset X$ we define

$\mu^*(A) = \inf \left\{ \sum_{i=1}^{\infty} p(E_j) \mid E_j \in E \text{ and } A \subseteq \bigcup_{j=1}^{\infty} E_j \right\}$

Then $\mu^*$ is an outer measure.

Proof: (a) Observe a countable collection of empty sets covers $A = \emptyset$ hence

$$\mu^*\emptyset = \inf \sum_{n \in \mathbb{N}} \rho(E_n) \leq \sum_{n \in \mathbb{N}} \rho(\emptyset) = 0$$

Also $\rho(E_n) \geq 0$ for any $E_n \in \mathcal{E}$ hence

$$\sum \rho(E_n) \geq 0$$

for any collection of intervals therefore

$$\inf \sum_{E_n \in \mathcal{E}} \rho(E_n) \geq 0.$$

So we get $\mu^*\emptyset = 0$.

For any non-empty set $A \subset X$ there exists $\{E_j\}_{j=1}^{\infty} \subset \mathcal{E}$ such that $A \subseteq \bigcup_{j=1}^{\infty} E_j$ we can set $E_j = X$ for all $j$.

(cb) If $A \subseteq B$ then if $\{E_j\}$ is collection of sets covering $B$ then $\mathfrak{B}$ covers $A$ as well.

If $\mathfrak{B}$ is a cover of $B$. We construct

$$A = \{E_n \in \mathfrak{B} \mid B_n \cap A \neq \emptyset\}$$

and observe $A \subseteq \mathfrak{B}$ Hence

$$\sum_{A_n \in A} P(A_n) \leq \sum_{B_n \in \mathfrak{B}} P(B_n)$$

therefore $\mu^*A \leq \sum_{B_n \in \mathfrak{B}} P(B_n)$. Observe $\mu^*A \leq \mu^*B$ otherwise $\mu^*A > \mu^*B$ then there exist a countable collection $\{E_j\}^2 = E$ covering $B$ and also $A$ whose sum i.e. $\sum P(B_n)$ is less than $\mu^*A$ which contradicts the fact that $\mu^*A$ is the infimum of sum of countable collection of sets in $E$ covering $A$.

(c) Suppose $\{A_j\}_{j=1}^{\infty} \subset \mathcal{P}(X)$ and fix $\delta > 0$. For each $j \in \mathbb{N}$ we get $\{E_{j,k}\}_{k=1}^{\infty} \subset \mathcal{E}$ such that

$$A_j \subset \bigcup_{k=1}^{\infty} E_{j,k} \quad \text{and} \quad \sum_{k=1}^{\infty} \rho(E_{j,k}) \leq \mu^*(A_j) + \frac{\delta}{2^j}.$$

But if $A = \bigcup_{j=1}^{\infty} A_j$, we have $A \subset \bigcup_{j,k=1}^{\infty} E_{j,k}$ and

$$\sum_{j,k} \rho(E_{j,k}) \leq \sum_{j} \rho(E_{j,k}) + \delta, \quad \text{whence}$$

$$\mu^*(A) \leq \sum_{j} \mu^*(A_j) + \delta. \quad \text{As } \epsilon \text{ is arbitrary, we are done.}$$

□

* A set $A$ is measurable if for each set $E$ we have $m^*E = m^*(A \cap E) + m^*(A^c \cap E)$

Note: We know for any set $m^*E \leq m^*(A \cap E) + m^*(A^c \cap E)$ holds. Hence $A$ is measurable iff

$$m^*E \geq m^*(A \cap E) + m^*(A^c \cap E) \quad \text{for any set } E.$$

*Remark* 0.2. If $A$ is measurable then $A^c$ is measurable.

**Lemma 0.3.** If $A_1$ and $A_2$ are measurable sets then $A_1 \cup A_2$ is measurable.

*Proof.* For any set $E \subset X$

$$\mu^*(E \cap (A_1 \cup A_2)) = \mu^*(E \cap A_1) + \mu^*(E \cap A_2 \cap A_1^c)$$

and

$$\mu^*(E \cap (A_1 \cup A_2)) + \mu^*(E \cap (A_1 \cup A_2)^c) = \mu^*(E \cap A_1) + \mu^*(E \cap A_2 \cap A_1^c) + \mu^*(E \cap A_2^c \cap A_1^c)$$

Now apply the fact that $A_2$ is a measurable set,

$$\mu^*(E \cap (A_1 \cup A_2)) + \mu^*(E \cap (A_1 \cup A_2)^c) = \mu^*(E \cap A_1) + \mu^*(E \cap A_1^c) = \mu^*(E)$$

as $A_1$ is a measurable set. Hence $A_1 \cup A_2$ is measurable. □

**Lemma 0.4.** Let $E$ be any set, and $A_1, \ldots, A_n$ a finite sequence of disjoint measurable sets. Then

$$\mu^*\left(E \cap \left(\bigcup_{i=1}^{n} A_i\right)\right) = \sum_{i=1}^{n} \mu^*(E \cap A_i)$$

*Proof.* Base case holds for $n = 1$. Assume the given holds for $n-1$ sets.

Observe $E \cap \left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n = E \cap A_n$ and

$$E \cap \left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n^c = E \cap \bigcup_{i=1}^{n-1} A_i$$

as $A_n$ is disjoint from $\xi_{i=1}^{n-1} A_i$. Hence

$$\mu^*\left(E \cap \left(\bigcup_{i=1}^{n} A_i\right)\right) =$$

$$\mu^*\left(E \cap \bigcup_{i=1}^{n-1} A_i \cap A_n\right) + \mu^*\left(E \cap \bigcup_{i=1}^{n-1} A_i \cap A_n^c\right)$$

by measurability of $A_n$. This implies

$$\mu^*\left(E \cap \left(\bigcup_{i=1}^{n} A_i\right)\right) = \mu^*(E \cap A_n) + \mu^*\left(E \cap \bigcup_{i=1}^{n-1} A_i\right)$$

□

By induction hypothesis's, we get

$$\mu^*\left(E \cap \left(\bigcup_{i=1}^{n} A_i\right)\right) = \sum_{i=1}^{n} \mu^*(E \cap A_i).$$

□

**Theorem 0.2.** (Carathéodory's Theorem) If $\mu^*$ is an outer measure on $X$, the collection $\mathcal{M}$ of $\mu^*$-measurable sets is a $\sigma$-algebra and the restriction of $\mu^*$ to $\mathcal{M}$ is a complete measure.

*Proof.* According to a previous remark and lemma, $\mathcal{M}$ is an algebra and finitely additive. As per the previous lemma,

$$\mu^*\left(E \cap \left(\bigcup_{i=1}^{\infty} A_i\right)\right) = \sum_{i=1}^{\infty} \mu^*(E \cap A_i).$$

holds for any $E \subset X$ and collection of $\mu^*$-measurable disjoint sets $\{A_i\}_{i=1}^{\infty}$. □

In order to $M$ is closed under disjoint unions.

Let $B_n = \bigcup_{j=1}^{n} A_j$ and $B = \bigcup_{j=1}^{\infty} A_j$. Hence

$$\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c)$$

$$\geq \sum_{j=1}^{n} \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$$

observe the statement above holds for any arbitrary $n$.

$$\mu^*(E) \geq \sum_{j=1}^{\infty} \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$$

$$\geq \mu^*\left(\bigcup_{j=1}^{\infty}(E\cap A_j)\right)+\mu^*(E\cap B^c)$$

$$= \mu^*(E\cap B)+\mu^*(E\cap B^c)\geq \mu^*(E).$$

It follows that $B\in M$ and $\mu^*(B)=\sum_{j=1}^{\infty}\mu^*(A_j)$, so $\mu^*$ is countably additive, by substituting $E=B$.

Finally, if $\mu^*(A)<\infty$ for any $E\subset X$ we have

$$\mu^*(E)\leq \mu^*(E\cap A)+\mu^*(E\cap A^c)$$

$$= \mu^*(E\cap A^c)\leq \mu(E).$$

so, $A\in\mathcal{M}$. Hence $\mu^*|_{\mathcal{M}}$ is complete measure

* Let $A\subset\mathcal{P}(X)$ is an algebra, a function $\mu_0 : A\to[0,\infty]$ will be called a premeasure if * $\mu_0(\emptyset)=0$, * if $\{A_j\}_{j\in\mathbb{N}}$ is a sequence of disjoint sets in $A$, then $\mu_0(\bigcup_{j=1}^{\infty}A_j)=\sum_{j=1}^{\infty}\mu_0(A_j)$.

**Note:** The difference between a measure and a premeasure is the domain.

* Measure is defined on a $\sigma$-algebra * Premeasure is defined on an algebra.

* We have a notion of finite and $\sigma$-finite premeasures as same as that of a measure.

*Remark* 0.3. If $\mu_0$ is a premeasure on $\mathcal{A}\subseteq\mathcal{PCX}$, it induces an outer measure on $X$ in accordance with Proposition /ref , namely

$$\mu^*(E)=\inf\left\{\sum_{j=1}^{\infty}\mu_0(A_j)\mid A_j\in\mathcal{A},\, E\subseteq\bigcup_{j=1}^{\infty}A_j\right\}$$

**Lemma 0.5.** If $\mu_0$ is a premeasure on $\mathcal{A}$ and $\mu^*$ as defined in remark above, then

1. $\mu^*|_{\mathcal{A}}=\mu_0$;
2. every set in $\mathcal{A}$ is $\mu^*$-measurable.

*Proof.* a) Suppose $E\in\mathcal{A}$. If $E\subseteq\bigcup_{j=1}^{\infty}A_j$ with any countable collection $\{A_j\}_{j=1}^{\infty}\subseteq\mathcal{A}$ then we let $B_n=E\cap\left(A_n\setminus\bigcup_{j=1}^{n-1}A_j\right)$. Observe $B_n$ is a disjoint member of $\mathcal{A}$ whose union is $E$, so □

$\mu_0(E)=\sum_{n=1}^{\infty}\mu_0(B_n)\leq\sum_{n=1}^{\infty}\mu_0(A_j)$. Since this holds for any arbitrary cover of $E$ hence as per the definition of $\mu^*$, we get $\mu_0(E)\leq\mu^*(E)$. Now as $E\in A$, we can choose $A_j\subset\bigcup_{j=1}^{\infty}A_j$ st. $A_1=E$ and $A_j=\phi_n$. Hence $\mu_0(E)=\mu^*(E)$.

b) If $A\in\mathcal{A}$, $E\subset X$ and $\epsilon>0$ then there is a sequence $B_j\subset_{j\in\mathbb{N}}\subset A$ with $E\subset\bigcup_{j=1}^{\infty}B_j$ and $\sum_{j=1}^{\infty}\mu_0(B_j)\leq\mu^*(E)+\epsilon$. Since $\mu_0$ is additive on $\mathcal{A}$,

$\mu_0(\bigcup_{j=1}^{\infty}B_j\cap X)=\mu_0(\bigcup_{j=1}^{\infty}B_j\cap(A\cup A^c))$

$= \mu_0(\bigcup_{j=1}^{\infty}(B_j \cap A)) + \mu_0(\bigcup_{j=1}^{\infty}(B_j \cap A^c))$

$= \sum_{j=1}^{\infty} \mu_0(B_j \cap A) + \sum_{j=1}^{\infty} \mu_0(B_j \cap A^c)$.

Hence,

$\mu^*(E) + \epsilon \geq \sum_{j=1}^{\infty} \mu_0(B_j \cap A) + \sum_{j=1}^{\infty} \mu_0(B_j \cap A^c)$

$\geq \mu_0(E \cap A) + \mu_0(E \cap A^c)$

As $\epsilon$ is arbitrary, $A$ is $\mu^*$-measurable.

**Theorem 0.3.** Let $A \subseteq \mathcal{P}(X)$ be an algebra, $\mu_0$ is a premeasure defined on $A$. Let $\mathcal{M}$ be $\sigma$-algebra generated by $A$. There exists a measure $\mu$ on $\mathcal{M}$ such that $\mu|_A = \mu_0$.

*Proof.* If we define $\mu^*$ as defined earlier and $\mu := \mu^*|_{\mathcal{M}}$. Then we know by the previous lemma $\mu^*|_A = \mu|_A = \mu_0$ and all $A$ are $\mu^*$-measurable. This implies $\sigma$-algebra of $\mu^*$-measurable sets include $A$. □

**Theorem 0.4.** If $\nu$ is another measure defined on $\mathcal{M}$ (as per the previous theorem) that extends on $\mu_0$ then $\nu(E) \leq \mu(E)$ for any $E \in \mathcal{M}$ and equality when $\mu(E) < +\infty$. If $\mu_0$ is $\sigma$-finite, then $\mu$ is the unique extension of $\mu_0$ to a measure on $\mathcal{M}$.

*Proof.* If $E \in \mathcal{M}$ and $E \subset \bigcup_{j=1}^{\infty} A_j$ where $A_j \in \mathcal{A}$ for all $j \in \mathbb{N}$, then $\nu(CE) \leq \sum_{j=1}^{\infty} \nu(A_j) = \sum_{j=1}^{\infty} \mu_0(A_j)$ holds true for any $\{A_j\}_{j=1}^{\infty}$ collection. Hence,

$$\nu(E) \leq \mu(CE).$$

Let $A = \bigcup_{j=1}^{\infty} A_j$. Now $\nu(A) = \nu\left(\bigcup_{j=1}^{\infty} A_j\right)$

$$= \sum_{j=1}^{\infty} \nu(A_j) = \lim_{n\to\infty} \sum_{j=1}^{n} \nu(A_j) = \lim_{n\to\infty} \nu\left(\bigcup_{j=1}^{n} A_j\right)$$

$$= \lim_{n\to\infty} \mu_0\left(\bigcup_{j=1}^{n} A_j\right) = \sum_{j=1}^{\infty} \mu_0(A_j) = \mu_0(A).$$

□

If $\mu(E) < +\infty$. For every $\epsilon > 0$ there exist a collection of sets $\{A_\epsilon, j\}_{j=1}^{\infty}$, such that $A_\epsilon = \bigcup_{j=1}^{\infty} A_{\epsilon j}$ such that $\mu(A_\epsilon) < \mu(CE) + \epsilon$, hence $\mu(A_\epsilon \setminus E) < \epsilon$, and $\mu(CE) \leq \mu(A) = \nu(A) = \nu(E) + \nu(A \setminus E) \leq \nu(CE) + \mu(A \setminus E) \leq \nu(CE) + \epsilon$ holds for all $\epsilon > 0$. This implies $\mu(CE) \leq \nu(CE)$.

Finally suppose $X = \bigcup_j A_j$ with $\mu_0(A_j) < \infty$ where we can assume $A_j$'s are disjoint (or reconstruct a given collection into that way). Then for any $E \in \mathcal{M}$

$\mu(E) = \sum_j \mu(LE \cap A_j) = \sum_j \nu(CE \cap A_j)$

$= \nu(CE)$,

so $\mu = \nu$.

A: finite disjoint unions of h-intervals is an algebra.

Borel Measure on $\mathbb{R}$

Notation: $(a, b]$ are denoted as h-intervals.

$\mu$ is the borel measure on $\mathcal{B}_{\mathbb{R}}$. We define distribution function as $F(x) = \mu((-\infty, x])$

Proposition

Let $F : \mathbb{R} \to \mathbb{R}$ be an increasing and right cont. If $\{(a_j, b_j]\}_{j=1}^n$ are disjoint h-intervals, let

$$\mu_0\left(\bigcup_{j=1}^n (a_j, b_j]\right) = \sum_{j=1}^n \mu_0((a_j, b_j]),$$

and let $\mu_0(\emptyset) = 0$. Then $\mu_0$ is a premeasure on algebra $A$.

Proof: (a) $\mu_0$ is well defined

Given any collection of disjoint h-intervals. If $\{(a_j, b_j]\}_{j=1}^n$ by relabelling, we get

$$a = a_1 < b_1 = a_2 < \cdots < b_n = b.$$

This implies $\sum_{j=1}^n [F(b_j) - F(a_j)] = F(b) - F(a)$.

More generally, if $\{I_i\}_{i=1}^n$ and $\{J_j\}_{j=1}^m$ are finite sequences of disjoint $h$-intervals such that $\bigcup_{i=1}^n I_i = \bigcup_{j=1}^m J_j$ by the reasoning above we can say,

$$\sum_i \mu_0(I_i) = \sum_{i,j} \mu_0(I_i \cap J_j) = \sum_j \mu_0(J_j).$$

(b) $\mu_0$ is countably additive

Observe $\mu_0$ is finitely additive by construction.

We want to show if $\sum I_i \subseteq \mathcal{U}$ st. $\bigcup_{i \in \mathbb{N}} I_i \in \mathcal{U}$ then

$$\mu_0\left(\bigcup_{i=1}^\infty I_i\right) = \sum_{i=1}^\infty \mu_0(I_i).$$

As $\bigcup_{i=1}^\infty I_i \in \mathcal{U}$, we can separate $\{I_i\}_{i=1}^\infty$ into finitely many subsequences whose union is an $h$-interval.

Assume $\bigcup_{i=1}^\infty I_i$ is an $h$-interval (otherwise you look into a smaller collection of $I_i$'s whose union is an $h$-interval.)

$I = \bigcup_{i=1}^\infty I_i = [a, b)$. Observe

$$\mu_0(I) = \mu_0\left(\bigcup_{j=1}^n I_j\right) + \mu_0\left(I \setminus \bigcup_{j=1}^n I_j\right)$$

$$\geq \mu_0\left(\bigcup_{j=1}^{n} I_j\right) = \sum_{j=1}^{n} \mu_0(I_j)$$

holds for any $n \in \mathbb{N}$, we obtain $\mu_0(I) \geq \sum_{j=1}^{n} \mu(I_j)$.

To show reverse inequality

case (ci): If $a, b$ are finite.

As $F$ is right continuous, for any $\epsilon > 0$, there exists $\delta_\epsilon > 0$ such that $F(a + \delta_\epsilon) - F(a) < \epsilon$.

If $I_j = (a_j, b_j]$ $j \in \mathbb{N}$ then we can find a $\delta_{\epsilon,j} > 0$ such that $F(b_j + \delta_{\epsilon,j}) - F(b_j) < \frac{\epsilon}{2^j}$.

Now $\{(a_j, b_j + \delta_{\epsilon,j})\}_{j=1}^{\infty}$ is a collection of open intervals covering $[a + \delta_\epsilon, b]$, so there is a finite subcover.

Let $(a_1, b_1 + \delta_1), \ldots, (a_N, b_N + \delta_N)$ cover $[a + \delta_\epsilon, b]$ and $b_i + \delta_i \in (a_{i+1}, b_{i+1} + \delta_{i+1})$.

(We have replaced $\delta_{\epsilon,i} = \delta_i$ for all $i = \{1, \ldots, N-1\}$.

For any $\epsilon > 0$, observe

$$\mu_0(I) < F(b) - F(a + \delta_\epsilon) + \epsilon$$

$$\leq F(b_N + \delta_N) - F(a_1) + \epsilon$$

$$= F\left(b_N + \delta_N\right) - F\left(a_N\right) + \sum_{i=1}^{N-1}\left[F\left(a_{j+1}\right) - F\left(a_j\right)\right] + \varepsilon$$

$$\leq F\left(b_N + \delta_N\right) - F\left(a_N\right) + \sum_{i=1}^{N}\left[F\left(b_j + \delta_j\right) - F\left(a_j\right)\right] + \varepsilon$$

$$< \left(\sum_{i=1}^{N} F\left(b_j\right) + \frac{\varepsilon}{2^j} - F\left(a_j\right)\right) + \varepsilon < \sum_{i=1}^{\infty} \mu(I_j) + 2\varepsilon.$$

As the above statement holds for all $\varepsilon > 0$, we are done.

**case (ii):** If one of them is NOT finite.

Say $a = -\infty$. Fix $\varepsilon > 0$.

Then for any $M > 0$, we can find an $t$ open cover $(a_j + b_j + \delta_j)$ covering $[-M, b]$. Hence by the same reasoning we get

$$F(b) - F(-M) \leq \sum_{j=1}^{\infty} \mu_0(I_j) + 2\varepsilon,$$

whereas if $b = \infty$, for any $M < \infty$ we get

$$F(M) - F(a) \leq \sum_{j=1}^{\infty} \mu_0(I_j) + 2\varepsilon.$$

We get the desired result by $\varepsilon \to 0$ followed by $M \to \infty$. (Go through this).

**Theorem 0.5.** If $F : \mathbb{R} \to \mathbb{R}$ is any increasing, right continuous $d^m$, there is a unique Borel measure $\mu_F$ on $\mathbb{R}$ such that $\mu_F((a, b]) = F(b) - F(a)$ for all $a, b$. If $G$ is another function similar to this then $G - F$ is constant iff $\mu_F = \mu_G$

Conversely if $\mu$ is a Borel measure on $\mathbb{R}$ that is finite on all bounded Borel sets and we define

$$F(x) = \begin{cases} \mu((0, x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((x, 0]) & \text{if } x < 0 \end{cases}$$

*Proof.* Each $F$ induces a premeasure on $\mathcal{U}$ by previous proposition. It is clear that $F$ and $G$ induce the same premeasure iff $F - G$ is constant, and these premeasure are $\sigma$-finite as $\mathbb{R} = \bigcup_{j \in \mathbb{Z}} (j, j+1]$. $\square$

The first two assertions therefore follow from Theorem. As for the converse, the monotonicity of $\mu$ implies $F$ to be an increasing function.

Observe $\mu$ is continuous from above and below implies the right continuity of $F$ for $x \geq 0$ and $x < 0$. It is evident that $\mu = \mu_F$ on $\mathcal{U}$, and hence $\mu = \mu_F$ on $\mathcal{B}_{\mathbb{R}}$ by uniqueness of Theorem.

*Remark* 0.4. ∘ The theory could have also been developed for $[a, b)$ intervals and $F$ as left cont. $S^n$.

∘ If $\mu$ is a finite Borel measure on $\mathbb{R}$ then $F(x) = \mu((-\infty, x])$ is the cumulative distribution function.

∘ Any increasing right cont. $F^n$ gives us a complete measure.

Given an increasing, right continuous function $F$. We get $(IR, \mathcal{M}_\mu, \mu)$ as a complete measure space where for any $E \in \mathcal{M}_\mu$,

$$\mu(E) = \inf \left\{ \sum_{j=1}^{\infty} [F(b_j) - F(a_j)] \mid E \subset \bigcup_{j=1}^{\infty} (a_j, b_j] \right\}$$

**Lemma 0.6.** For any $E \in \mathcal{M}_\mu$,

$$\mu(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu((a_j, b_j)) \mid E \subset \bigcup_{j=1}^{\infty} (a_j, b_j] \right\}$$

*Proof.* Let $\nu : \mathcal{M}_\mu \to [0, \infty)$ such that

$$\nu(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu((a_j, b_j)) \mid E \subset \bigcup_{j=1}^{\infty} (a_j, b_j] \right\}$$

and $\mu$ be the measure defined in above given def$^n$.

Goal : $\nu(E) = \mu(E)$ for all $E \in \mathcal{M}_\mu$. $\square$

(i) We want to show $\mu(E) \le \nu(E)$ for any $E \in \mathcal{M}_\mu$.

Suppose $E \subseteq \bigcup_{j=1}^{\infty}(a_j, b_j)$ and for each $(a_j, b_j)$ we get $\{I_j^k\}_{k=1}^{\infty}$ as disjoint $h$-intervals covering $(a_j, b_j)$ for each $k$. Let $I_j^k = [c_j^k, c_j^{k+1}]$ where $c_j^1 = a_j$ and $c_j^k \to b_j$ as $k \to \infty$. Thus $E \subseteq \bigcup_{j,k} I_{j,k}$. Hence

$$\sum_{j=1}^{\infty} \mu((a_j, b_j)) = \sum_{j=1}^{\infty} \mu(I_j^k) \ge \mu(E)$$

(Or you can use $E \subseteq \bigcup_{j\in\mathbb{N}}(a_j, b_j)$ and by monotonicity and subadditivity and apply infimum, we get $\mu(E) \le \nu(E)$).

(ii) WTS : $\nu(E) \le \mu(E)$; for any $\varepsilon > 0$, we get $\{(a_j, b_j)\}_{j=1}^{\infty}$ covering $E$ and

$$\sum_{j=1}^{\infty} \mu((a_j, b_j)) \le \mu(E) + \varepsilon.$$

As $F$ is right cont. and increasing, for $\epsilon > 0$ we find $\delta_{\epsilon j} = \delta_j$ for each $j \in \mathbb{N}$ st.

$$F(b_j + \delta_j) - F(b_j) < \epsilon/2^j.$$

Then $E \subset \bigcup_{j=1}^{\infty}(a_j + b_j + \delta_j)$ and

$$\begin{aligned} \mu((a_j, b_j + \delta_j)) &\le \mu((a_j, b_j + \delta_j)) \\ &= F(b_j + \delta_j) - F(a_j) \\ &= F(b_j + \delta_j) - F(b_j) + F(b_j) - F(a_j) \end{aligned}$$

for all $j \in \mathbb{N}$. Now by taking a sum

$$\begin{aligned} \sum_{j=1}^{\infty} \mu((a_j, b_j)) &\le \sum_{j=1}^{\infty} F(b_j + \delta_j) - F(b_j) \\ &\quad + \sum_{j=1}^{\infty} \mu((a_j, b_j)) \\ &\le \epsilon + \epsilon + \mu(E). \end{aligned}$$

Observe $\nu(E) \le \sum_{j=1}^{\infty} \mu((a_j, b_j)) \le \mu(E) + 2\epsilon$ holds for all $\epsilon > 0$. Hence $\nu(E) \le \mu(E)$.

**Theorem 0.6.** If $E \in \mathcal{M}_\mu$, then

$$\begin{aligned} \mu(E) &= \inf \{\mu(U) \mid U \supseteq E \text{ and } U \text{ is open}\} \\ &= \sup \{\mu(K) \mid K \subseteq E \text{ and } K \text{ is compact}\} \end{aligned}$$